**Lab 1: Requirements Engineering & UML Use-Case Modelling**

**PROBLEM STATEMENT #35 | Retail, E-Commerce & Finance**

**Customizable Subscription Box Scheduler**

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**Section- G**

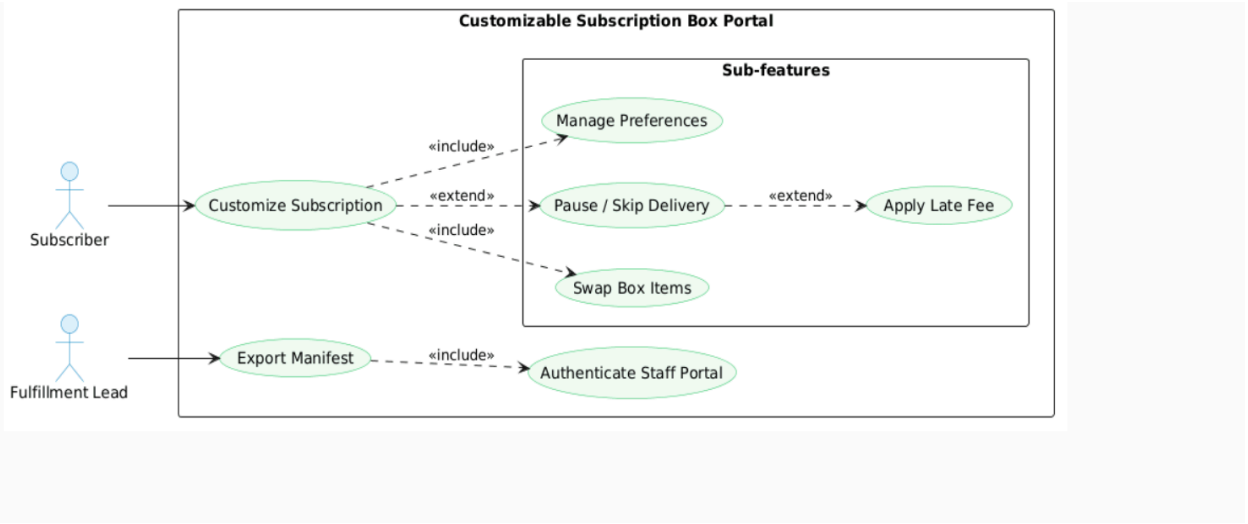
1) Functional and Non-Functional Requirements:

| ID | Type | Description | Priority | Acceptance Criteria | Rationale |
|---|---|---|---|---|---|
| FR-001 | Functional | The system shall allow subscribers to swap monthly box items or pause deliveries up to 48 hours prior to the monthly billing renewal date. | High | **Pass:** Selection updated and fulfillment manifest refreshed<br><br>**Fail:** Customizatio n locked out prematurely. | Ensures flexibility for users while giving operations enough lead time for packing. |
| FR-002 | Functional | The system shall enable subscribers to configure and store preference tags (e.g., genres, dietary restrictions, hobbies) during onboarding and profile updates. | High | **Pass:** Tags save successfully and dynamically filter available product catalogs.<br><br>**Fail:** Profile update fails to sync with recommendat ion engine. | Crucial for personalizing subscription contents to individual user preferences. |
| FR-003 | Functional | The system shall automatically generate recurring monthly invoices and execute automated payment processing via integrated payment | High | **Pass:** Invoice generated and payment charged successfully; receipt emailed.<br><br>**Fail:** Gateway | Automates revenue collection and maintains steady operational cycles. |

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| | | gateways on the designated billing date . | | timeout halts processing without error logging. | |
| FR-004 | Functional | The Fulfillment Lead shall be able to view, filter, and export the consolidated monthly fulfillment manifest grouped by delivery zone and product type. | Medium | **Pass:** CSV/PDF export downloads accurately for selected fulfillment cycles.<br><br>**Fail:** Export times out or omits active subscriber items. | Streamlines warehouse packaging and logistics coordination . |
| FR-005 | Functional | The system shall notify subscribers via email and push notifications 72 hours prior to the upcoming billing cycle lock window. | Medium | **Pass:** Notification dispatched within 5 minutes of schedule trigger.<br><br>**Fail:** Notification fails to queue due to invalid user contact states. | Prevents accidental unwanted renewals and enhances customer trust. |
| NFR-001 | Performance & Security | The fulfillment manifest generator | High | **Pass:** Benchmarkin g tests confirm target | Guarantees system scalability ahead of |

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| | | must export shipping labels for 10,000 monthly boxes in under 1 minute. | | latency and security standards under simulated peak load.<br><br>**Fail:** Latency exceeds 60 seconds during stress tests. | high-volume seasonal spikes (e.g., holidays). |
| NFR-002 | Security & Compliance | All subscriber payment profiles, shipping addresses, and personal preference data must be encrypted at rest and in transit using industry-standard AES-256 and TLS 1.3 protocols. | High | **Pass:** Security audit and vulnerability scans report zero unencrypted sensitive payloads.<br><br>**Fail:** Plaintext data detected in database logs. | Protects sensitive consumer financial data and ensures regulatory compliance. |

2. UML Use Case Diagram:



3. Use-Case Flow Specification:

Preconditions:

- Subscriber is authenticated and logged into the subscription portal.
- The customer account has an active, paid subscription status for the current billing cycle.
- The current system timestamp is strictly greater than 48 hours prior to the scheduled monthly billing renewal date.

Postconditions (on success)

- Subscription database profile is successfully updated with the subscriber's new item selections and preference tags.
- Operational fulfillment manifest is refreshed to reflect the new box contents.
- Automated confirmation email detailing the updated subscription parameters is dispatched to the subscriber.

Main Success Scenario

- Subscriber navigates to the "My Subscription" dashboard.
- System retrieves and displays current box contents, applied preference tags, and the upcoming billing renewal date.
- Subscriber selects an item from the current box manifest and chooses to swap it with an alternative catalog item matching their preference tags.
- System validates inventory stock availability for the replacement item and temporarily locks the choice in the staging database.
- Alternatively, subscriber selects the option to pause deliveries or skip the upcoming billing cycle.
- System «include» invokes Calculate Next Billing Date (UC-03) to adjust the recurring schedule timeline.
- Subscriber confirms the modifications and clicks "Save Changes".
- System updates the live fulfillment manifest and commits the database transaction.
- Subscriber receives an email confirmation with a summary of the updated delivery box.
- Use case ends successfully.

Alternate Flow – 5a. 48-Hour Modification Lockout Window

- Subscriber attempts to execute an item swap or delivery pause within the 48-hour window preceding the monthly billing renewal date.
- System detects that the time remaining until billing is less than or equal to 48 hours.
- System locks out customization controls and blocks the "Save Changes" action.
- System prompts the subscriber with a warning message stating that the customization window is closed and changes cannot be applied for the upcoming cycle.
- System redirects the user interface to view settings for the subsequent billing cycle. Use case ends in failure.